(4) Install air cleaner tube to throttle body.

4.0L

(1) Install IAC motor to throttle body.
(2) Install and tighten two mounting bolts (screws) to 7 N·m (60 in. lbs.) torque.
(3) Install electrical connector.
(4) Install air cleaner tube to throttle body.

INTAKE AIR TEMPERATURE SENSOR

DESCRIPTION

The 2–wire Intake Manifold Air Temperature (IAT) sensor is installed in the intake manifold with the sensor element extending into the air stream.

The IAT sensor is a two-wire Negative Thermal Coefficient (NTC) sensor. Meaning, as intake manifold temperature increases, resistance (voltage) in the sensor decreases. As temperature decreases, resistance (voltage) in the sensor increases.

OPERATION

The IAT sensor provides an input voltage to the Powertrain Control Module (PCM) indicating the density of the air entering the intake manifold based upon intake manifold temperature. At key-on, a 5–volt power circuit is supplied to the sensor from the PCM. The sensor is grounded at the PCM through a low-noise, sensor-return circuit.

The PCM uses this input to calculate the following:

- Injector pulse-width
- Adjustment of spark timing (to help prevent spark knock with high intake manifold air-charge temperatures)

The resistance values of the IAT sensor is the same as for the Engine Coolant Temperature (ECT) sensor.

REMOVAL

2.4L

The intake manifold air temperature (IAT) sensor is installed into the intake manifold plenum at the rear end of the intake manifold (Fig. 15).

(1) Disconnect electrical connector from IAT sensor.
(2) Clean dirt from intake manifold at sensor base.
(3) Gently lift on small plastic release tab (Fig. 15) or (Fig. 16) and rotate sensor about 1/4 turn counterclockwise for removal.
(4) Check condition of sensor o-ring (Fig. 16).

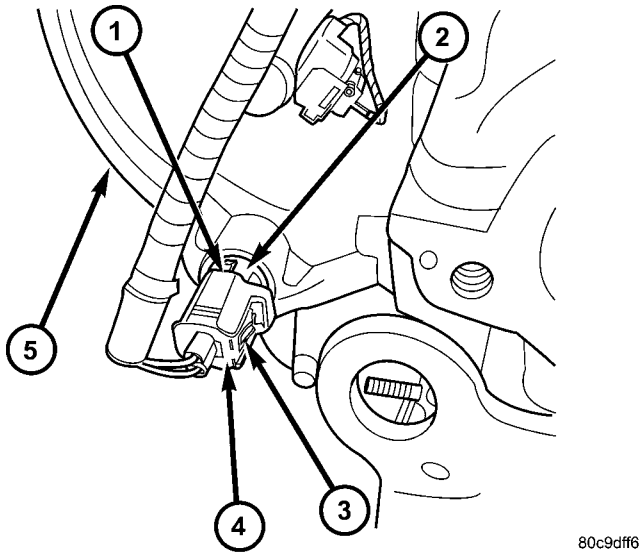

Fig. 15 IAT SENSOR - 2.4L

1 - RELEASE TAB
2 - IAT SENSOR
3 - PRESS HERE FOR REMOVAL
4 - ELECTRICAL CONNECTOR
5 - REAR END OF INTAKE MANIFOLD

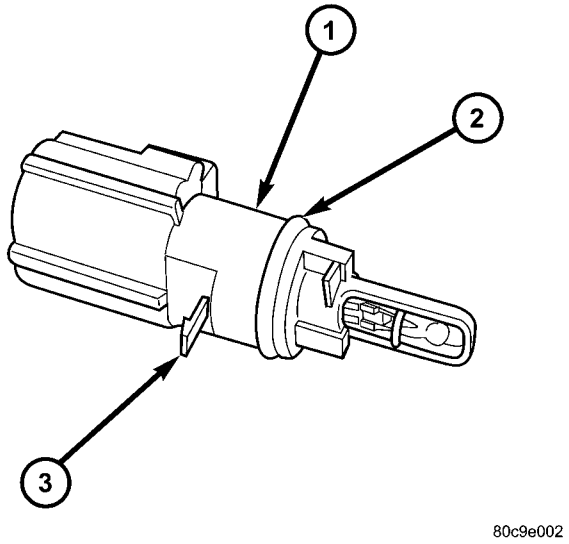

Fig. 16 IAT SENSOR TAB / O-RING - 2.4L

1 - IAT SENSOR
2 - SENSOR O-RING
3 - RELEASE TAB

4.0L

The intake manifold air temperature (IAT) sensor is installed into intake manifold plenum near throttle body (Fig. 17).